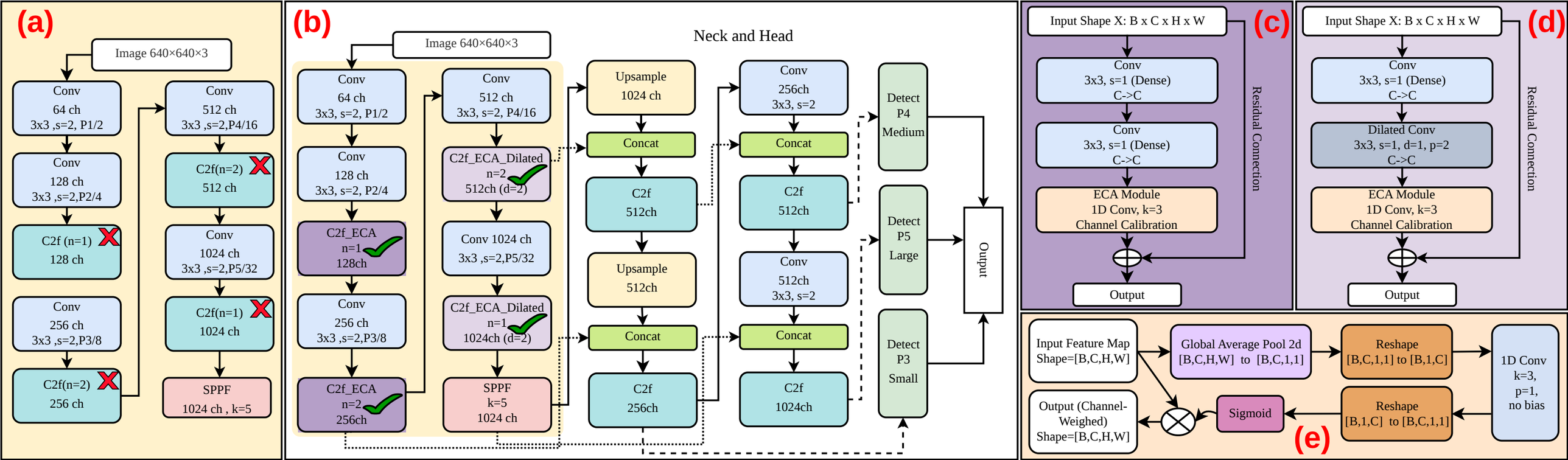

(a)
Image 640×640×3
Conv
64 ch
3x3 ,s=2, P1/2
Conv
128 ch
3x3 ,s=2,P2/4
C2f (n=1)
128 ch
Conv
256 ch
3x3 ,s=2,P3/8
C2f(n=2)
256 ch
Conv
512 ch
3x3 ,s=2,P4/16
C2f(n=2)
512 ch
Conv
1024 ch
3x3 ,s=2,P5/32
C2f(n=1)
1024 ch
SPPF
1024 ch , k=5
(b)
Neck and Head
Image 640×640×3
Conv
64 ch
3x3, s=2, P1/2
Conv
128 ch
3x3, s=2, P2/4
C2f_ECA
n=1
128ch
Conv
256 ch
3x3 ,s=2,P3/8
C2f_ECA
n=2
256ch
Conv
512 ch
3x3, s=2, P4/16
C2f_ECA_Dilated
n=2
512ch (d=2)
Conv 1024 ch
3x3 ,s=2,P5/32
C2f_ECA_Dilated
n=1
1024ch (d=2)
SPPF
k=5
1024 ch
Upsample
1024 ch
Concat
C2f
512ch
Upsample
512ch
Concat
C2f
256ch
Conv
256ch
3x3, s=2
Concat
C2f
512ch
Conv
512ch
3x3, s=2
Concat
C2f
1024ch
Detect
P4
Medium
Detect
P5
Large
Detect
P3
Small
Output
(c)
Input Shape X: B x C x H x W
Conv
3x3, s=1 (Dense)
C->C
Conv
3x3, s=1 (Dense)
C->C
ECA Module
1D Conv, k=3
Channel Calibration
Output
Residual Connection
(d)
Input Shape X: B x C x H x W
Conv
3x3, s=1 (Dense)
C->C
Dilated Conv
3x3, s=1, d=1, p=2
C->C
ECA Module
1D Conv, k=3
Channel Calibration
Output
Residual Connection
(e)
Input Feature Map
Shape=[B,C,H,W]
Global Average Pool 2d
[B,C,H,W] to [B,C,1,1]
Reshape
[B,C,1,1] to [B,1,C]
1D Conv
k=3,
p=1,
no bias
Reshape
[B,1,C] to [B,C,1,1]
Sigmoid
Output (Channel-Weighed)
Shape=[B,C,H,W]